CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating & Evidence-Grounded Reasoning for Trustworthy Fake News Triage

AUDIT ID: AUD-219586
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FRAMEWORK: CAGE-Fake v2.0

LEGAL & FACT-CHECKING POLICY: A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify factual reality. URL extraction, pattern recognition, and external evidence verification remain strictly separate processes. The pipeline abstains (**UNCERTAIN**) whenever evidence is missing, conflicting, or model consensus is low.

Title: Senate Approves Major Clean Energy Infrastructure Bill
Source: https://example.com/senate-clean-energy
Word Count: 450 words | **Extraction:** OK
OOD Detector: Loaded (SBERT & Vocab) | **Triage Policy:** Evidence-Grounded Three-Way Gating

TRIAGE VERDICT: REAL
Calibrated Confidence: 94.2% (τ = 80.0%)
Model Disagreement (std): 0.035 (δ = 0.20)
Safety Trigger: Verified Evidence & Model Consensus

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Vote	Confidence
Random Forest	8.0%	REAL	92.0%
XGBoost	5.0%	REAL	95.0%
LightGBM	6.0%	REAL	94.0%
Logistic Regression	9.0%	REAL	91.0%
CAGE-Fake Stack	5.8%	REAL	94.2%

Consensus Status: Unanimous Consensus
Vote Distribution: **0 FAKE** vs **5 REAL**
Base Disagreement (4-base std): 0.035 (δ = 0.20)
5-Model Consensus Spread: 0.015
[==----------]
Safety Assessment: All 5 models exhibit nearly identical probability outputs.

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	92.0/100	Excellent / Restrained
Source Attribution	85.0/100	Excellent / Restrained
Epistemic Grounding	88.0/100	Excellent / Restrained
Punctuation Discipline	95.0/100	Excellent / Restrained
Readability Structure	80.0/100	Excellent / Restrained
Lexical Diversity	78.0/100	Adequate / Neutral

OVERALL CREDIBILITY INDEX
86.8 / 100
Credibility Tier: A (Restrained / Professional Copy)

Scores are standardized against verified newsroom copy standards: high factual attribution, balanced syntax, and low sensationalism maximize the index.

3. Grounded Diagnostic Reasoning & Counterfactual Sensitivity

Diagnostic Explanation: The article presents well-attributed statements with high inter-model consensus.

Counterfactual Simulation: Article employs factual, restrained syntax. Original P(FAKE): **5.8%** → Transformed P(FAKE): **5.8%** **(Signed Shift: +0.0%)**

[v] Clean Stylometrics: No significant sensational exaggeration, clickbait patterns, or manipulative phrasing detected.

4. Claim-Level Semantic Evidence Grounding (Triage Verification)

No external claim-level verification data required for this article. Pipeline relied on linguistic stylometrics and calibrated model consensus.

5. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Impact	Top REAL-Driving Anchors	Impact
Lexical Diversity	+0.010	Journalistic Source Attribution Density	-0.250

6. Empirical Benchmark Model Comparison (Held-out Test Split)

Model Name	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest	0.8900	0.8900	0.0700	100.0%	0.8900
CAGE-Fake (Calibrated + Gated)	**0.9300**	**0.9300**	**0.0300**	**82.0%**	**0.9700**